Case Study 3: Retail Billing with Multiple Discount Rules

Aim: To develop a Python program that processes multiple products, calculates quantities, applies category-specific taxes, provides discounts based on purchase amount, enforces maximum concession limits, and generates an itemized invoice.

1. Problem Statement

A retail store sells different products under different categories. Each category has a specific tax rate, and customers receive discounts based on their total purchase amount.

The program should:

1. Accept multiple product details.

2. Enter product name, category, price, and quantity.

3. Calculate the total price of each product.

4. Apply category-specific taxes.

5. Apply discounts based on purchase thresholds.

6. Ensure the discount does not exceed the maximum concession limit.

7. Generate an itemized invoice.

8. Display the final bill amount.

9. Assumed Business Rules

Condition Rule

Electronics tax 18%

Clothing tax 5%

Groceries tax2%

Purchase below ₹1,000 No discount

Purchase ₹1,000–₹4,999 5% discount

Purchase ₹5,000–₹9,999 10% discount

Purchase ₹10,000 and above 15% discount

Maximum concession ₹2,000

For this example, the discount is applied to the subtotal before tax.

3. Algorithm

4. Start.

5. Enter the number of products.

6. Read product name, category, price, and quantity.

7. Calculate the subtotal of each product.

8. Calculate the total purchase amount.

9. Identify the discount percentage based on the purchase amount.

10. Calculate the discount.

11. Apply the maximum concession limit.

12. Calculate category-specific tax for each product.

13. Calculate the final bill.

14. Display the itemized invoice.

15. Stop.

16. Pseudocode

```
START

INPUT number_of_products

SET subtotal = 0
SET total_tax = 0

FOR each product

    INPUT product_name
    INPUT category
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    INPUT price
    INPUT quantity

    CALCULATE item_total = price * quantity

    ADD item_total to subtotal

    STORE product details

END FOR

IF subtotal >= 10000
    discount_rate = 15%
ELSE IF subtotal >= 5000
    discount_rate = 10%
ELSE IF subtotal >= 1000
    discount_rate = 5%
ELSE
    discount_rate = 0%
END IF

discount = subtotal * discount_rate

IF discount > 2000
    discount = 2000
END IF
```

FOR each product

CALCULATE category tax

ADD tax to total_tax

END FOR

final_bill = subtotal - discount + total_tax

DISPLAY itemized invoice

DISPLAY discount

DISPLAY total tax

DISPLAY final bill

STOP

5. Python Program

Complete Python solution

Retail Billing with Multiple Discount Rulesproducts = []subtotal = 0total_tax = 0# Category-specific tax ratestax_rates = { "electronics": 18, "clothing": 5, "groceries": 2}n = int(input("Enter number of products: "))# Accept product detailsfor i in range(n): print("\nProduct", i + 1) name = input("Enter product name: ") category = input("Enter category: ").lower() price = float(input("Enter price: ")) quantity = int(input("Enter quantity: ")) if price < 0 or quantity <= 0: print("Invalid price or quantity. Try again.") continue if category not in tax_rates: print("Invalid category. Product skipped.") continue

6. Sample Output

Enter number of products: 3

Product 1

Enter product name: Laptop

Enter category: electronics

Enter price: 50000

Enter quantity: 1

Product 2

Enter product name: Shirt

Enter category: clothing

Enter price: 1000

Enter quantity: 2

Product 3

Enter product name: Rice

Enter category: groceries

Enter price: 500

Enter quantity: 2

========== ITEMIZED INVOICE ==========

Product	Category	Qty	Price	Total	Tax
Laptop	electronics	1	50000	50000	9000
Shirt	clothing	2	1000	2000	100

Rice groceries 2 500 1000 20

Subtotal: Rs. 53000.0

Discount Rate: 15 %

Discount Amount: Rs. 2000

Maximum Concession: Rs. 2000

Total Tax: Rs. 9120.0

Final Bill Amount: Rs. 60120.0

========== THANK YOU! ==========

7. Flowchart

#chatgpt-mermaid-_r_238_{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;fill:rgb(237, 237, 237);}@keyframes edge-animation-frame{from{stroke-dashoffset:0;}}@keyframes dash{to{stroke-dashoffset:0;}}#chatgpt-mermaid-_r_238_ .edge-animation-slow{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 50s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_238_ .edge-animation-fast{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 20s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_238_ .error-icon{fill:rgb(48, 48, 48);}#chatgpt-mermaid-_r_238_ .error-text{fill:rgb(237, 237, 237);stroke:rgb(237, 237, 237);}#chatgpt-mermaid-_r_238_ .edge-thickness-normal{stroke-width:1px;}#chatgpt-mermaid-_r_238_ .edge-thickness-thick{stroke-width:3.5px;}#chatgpt-mermaid-_r_238_ .edge-pattern-solid{stroke-dasharray:0;}#chatgpt-mermaid-_r_238_ .edge-thickness-invisible{stroke-width:0;fill:none;}#chatgpt-mermaid-_r_238_ .edge-pattern-dashed{stroke-dasharray:3;}#chatgpt-mermaid-_r_238_ .edge-pattern-dotted{stroke-dasharray:2;}#chatgpt-mermaid-_r_238_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_238_ .marker.cross{stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_238_ svg{font-family:-apple-system-body,ui-sans-serif,-

apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;}#chatgpt-mermaid-_r_238_ p{margin:0;}#chatgpt-mermaid-_r_238_ .label{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_238_ .cluster-label text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_238_ .cluster-label span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_238_ .cluster-label span p{background-color:transparent;}#chatgpt-mermaid-_r_238_ .label text,#chatgpt-mermaid-_r_238_ span{fill:rgb(237, 237, 237);color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_238_ .node rect,#chatgpt-mermaid-_r_238_ .node circle,#chatgpt-mermaid-_r_238_ .node ellipse,#chatgpt-mermaid-_r_238_ .node polygon,#chatgpt-mermaid-_r_238_ .node path{fill:rgb(9, 23, 44);stroke:rgb(31, 78, 148);stroke-width:1px;}#chatgpt-mermaid-_r_238_ .rough-node .label text,#chatgpt-mermaid-_r_238_ .node .label text,#chatgpt-mermaid-_r_238_ .image-shape .label,#chatgpt-mermaid-_r_238_ .icon-shape .label{text-anchor:middle;}#chatgpt-mermaid-_r_238_ .node .katex path{fill:#000;stroke:#000;stroke-width:1px;}#chatgpt-mermaid-_r_238_ .rough-node .label,#chatgpt-mermaid-_r_238_ .node .label,#chatgpt-mermaid-_r_238_ .image-shape .label,#chatgpt-mermaid-_r_238_ .icon-shape .label{text-align:center;}#chatgpt-mermaid-_r_238_ .node.clickable{cursor:pointer;}#chatgpt-mermaid-_r_238_ .root .anchor path{fill:rgb(175, 175, 175)!important;stroke-width:0;stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_238_ .arrowheadPath{fill:rgb(175, 175, 175);}#chatgpt-mermaid-_r_238_ .edgePath .path{stroke:rgb(175, 175, 175);stroke-width:1px;}#chatgpt-mermaid-_r_238_ .flowchart-link{stroke:rgb(175, 175, 175);fill:none;}#chatgpt-mermaid-_r_238_ .edgeLabel{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_238_ .edgeLabel p{background-color:rgb(0, 0, 0);}#chatgpt-mermaid-_r_238_ .edgeLabel rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_238_ .labelBkg{background-color:rgba(0, 0, 0, 0.5);}#chatgpt-mermaid-_r_238_ .cluster rect{fill:rgb(48, 48, 48);stroke:rgba(255, 255, 255, 0.15);stroke-width:1px;}#chatgpt-mermaid-_r_238_ .cluster text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_238_ .cluster span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_238_ div.mermaidTooltip{position:absolute;text-align:center;max-width:200px;padding:2px;font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:12px;background:rgb(48, 48, 48);border:1px solid rgba(255, 255, 255, 0.15);border-radius:2px;pointer-events:none;z-index:100;}#chatgpt-mermaid-_r_238_ .flowchartTitleText{text-anchor:middle;font-size:18px;fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_238_ rect.text{fill:none;stroke-width:0;}#chatgpt-mermaid-_r_238_ .icon-shape,#chatgpt-mermaid-_r_238_ .image-shape{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_238_ .icon-shape p,#chatgpt-mermaid-

_r_238_ .image-shape p{background-color:rgb(0, 0, 0);padding:2px;}#chatgpt-mermaid-_r_238_ .icon-shape .label rect,#chatgpt-mermaid-_r_238_ .image-shape .label rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_238_ .label-icon{display:inline-block;height:1em;overflow:visible;vertical-align:-0.125em;}#chatgpt-mermaid-_r_238_ .node .label-icon path{fill:currentColor;stroke:revert;stroke-width:revert;}#chatgpt-mermaid-_r_238_ .node .neo-node{stroke:rgb(31, 78, 148);}#chatgpt-mermaid-_r_238_ [data-look="neo"].node rect,#chatgpt-mermaid-_r_238_ [data-look="neo"].cluster rect,#chatgpt-mermaid-_r_238_ [data-look="neo"].node polygon{stroke:url(#chatgpt-mermaid-_r_238_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_238_ [data-look="neo"].swimlane.cluster rect{filter:none;}#chatgpt-mermaid-_r_238_ [data-look="neo"].node path{stroke:url(#chatgpt-mermaid-_r_238_-gradient);stroke-width:1px;}#chatgpt-mermaid-_r_238_ [data-look="neo"].node .outer-path{filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_238_ [data-look="neo"].node .neo-line path{stroke:rgb(31, 78, 148);filter:none;}#chatgpt-mermaid-_r_238_ [data-look="neo"].node circle{stroke:url(#chatgpt-mermaid-_r_238_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_238_ [data-look="neo"].node circle .state-start{fill:#000000;}#chatgpt-mermaid-_r_238_ [data-look="neo"].icon-shape .icon{fill:url(#chatgpt-mermaid-_r_238_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_238_ [data-look="neo"].icon-shape .icon-neo path{stroke:url(#chatgpt-mermaid-_r_238_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_238_ .node text{font-size:14px;font-weight:600;letter-spacing:normal;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_238_ .edgeLabels text{font-size:13px;font-weight:600;letter-spacing:-0.08px;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_238_ .node tspan[font-weight="normal"],#chatgpt-mermaid-_r_238_ .edgeLabels tspan[font-weight="normal"]{font-weight:600;}#chatgpt-mermaid-_r_238_ .edgeLabel .label rect{opacity:1;rx:13px;ry:13px;fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-width:1px;}#chatgpt-mermaid-_r_238_ .node rect,#chatgpt-mermaid-_r_238_ .node circle,#chatgpt-mermaid-_r_238_ .node ellipse,#chatgpt-mermaid-_r_238_ .node polygon,#chatgpt-mermaid-_r_238_ .node path{fill:rgb(0, 40, 77);stroke:rgba(255, 255, 255, 0.1);stroke-width:1px;}#chatgpt-mermaid-_r_238_ .node rect{rx:16px;ry:16px;}#chatgpt-mermaid-_r_238_ .mermaid-decision .label-container{fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-dasharray:2,2;}#chatgpt-mermaid-_r_238_ .edgePaths .flowchart-link{stroke:rgb(175, 175, 175);stroke-width:1px;stroke-linecap:round;stroke-linejoin:round;}#chatgpt-mermaid-_r_238_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_238_ :root{--mermaid-font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";}StartEnter number of productsEnter product detailsCalculate item totalAdd

item total to subtotalMore products?Calculate discount rateCalculate discountDiscount exceeds 2000?Set discount to 2000Calculate category taxesCalculate final billDisplay itemized invoiceStopYesNoYesNo

8. Concepts Used from Syllabus

Syllabus concept	Application
Variables	Price, quantity, subtotal
Data types	Integer, float, string
Operators	Arithmetic and comparison
Conditional statements	Discount and tax rules
Loops	Processing multiple products
Strings	Product names and categories
Lists	Storing product records
Tuples	Maintaining product details
Nested data	Product and tax information
Algorithms	Billing calculations

9. Conclusion

The Retail Billing System calculates product prices, applies category-specific taxes, and provides discounts based on purchase value. It also enforces a maximum concession limit and generates an itemized invoice. This program demonstrates how Python can be used to automate billing operations in retail stores.

Note: The tax rates and discount percentages are assumed for this academic case study.